Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 13
Start date: 02-Feb-2026
Names:

S.no	Name
1	Boppana Anvitha
2	Sathyadev Selvam
3	Gautam Shivam Mishra
4	Shaik Firoz
5	Aayushi Maurya

Internship Duration: 8 Weeks

Project Title

Smart Electronics Inventra Management System

1. Project Objective

- Design and develop a centralized intelligent inventory management system tailored for electronics products to digitally monitor, control, and optimize stock handling processes
- Enable real-time inventory tracking to provide accurate visibility of electronics stock availability, movement, and usage patterns across product categories such as components, gadgets, and accessories
- Minimize human errors, stock mismatches, and data redundancy by replacing manual record-keeping with an automated digital solution for electronics inventory
- Implement role-based authentication and authorization to ensure secure and controlled access for administrators, managers, and staff
- Develop a transaction management module to record item issuance, returns, and stock updates with complete traceability for all electronics items
- Integrate email and SMS alert notifications to inform relevant users when stock levels fall below predefined thresholds for critical electronics components
- Provide Export to Excel and PDF functionality for inventory reports, transaction logs, and analytics summaries to support managerial decision-making

- Implement AI-driven future demand prediction to forecast stock requirements based on historical usage trends of electronics products

2. Project Description in Detail

Smart Electronics Inventra Management System is a smart digital platform developed to modernize and streamline traditional inventory handling processes specifically for the electronics domain. In many organizations, electronics inventory is still managed using manual registers or basic spreadsheets, which often results in stock mismatches, delayed updates, data redundancy, and lack of real-time visibility. This system addresses these challenges by providing a centralized, automated, and secure inventory management solution.

The system allows authorized users to manage electronics inventory items with complete control over stock records. Each product entry includes detailed attributes such as item name, category, quantity, supplier information, price, and stock status. Through an intuitive interface, users can easily add new items, update stock levels, delete outdated records, and search for products, ensuring efficient day-to-day electronics inventory operations.

The system incorporates a role-based authentication system to maintain data security and controlled access. Administrators can manage users and oversee the entire inventory, while staff members can perform permitted operations such as stock updates and transaction entries. This structure ensures both accountability and data protection within the system.

A core feature of the system is its transaction tracking module, which records all stock movements including item issuance and returns. Every transaction is logged with timestamps and user details, enabling full traceability and reducing the risk of stock discrepancies or misuse. The system automatically updates inventory quantities after each transaction, ensuring that stock information remains accurate and up to date.

To enhance decision-making, the system provides an interactive dashboard with data visualization tools. The dashboard displays key insights such as total stock, low-stock items, frequently used electronics products, and category-wise distribution. Visual elements like charts and summary cards help users quickly understand inventory trends and plan restocking efficiently.

Additionally, the system includes a smart alert mechanism that notifies users via email and SMS when stock levels fall below predefined thresholds. Users can export inventory data and transaction reports to Excel or PDF formats for offline analysis and reporting. A future demand prediction module uses historical transaction data to forecast upcoming stock requirements, enabling proactive procurement planning for electronics components. The system is designed with a scalable architecture, allowing it to be adapted for various environments such as electronics retail stores, warehouses, educational institutions, and corporate offices.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Requirement analysis, problem identification, and system architecture design for electronics domain	Conducted detailed study of existing electronics inventory challenges, finalized system requirements, and designed overall system architecture and module breakdown
Week 2	Database schema design and backend setup	Designed normalized relational MySQL database schema for electronics items, users, and transactions; configured Spring Boot

Week	Activities Planned	Activities Completed
		backend and API structure
Week 3	Authentication system and role-based access control	Implemented secure login system, user registration, password encryption, and role-based authorization for Admin and Staff users
Week 4	Inventory management module development	Developed core inventory features including item addition, stock updates, deletion, category classification for electronics, and search functionality
Week 5	Transaction management system and alert notifications	Implemented item issue and return tracking with automatic stock quantity adjustments; integrated email and SMS alert system for low-stock notifications
Week 6	Dashboard development, data visualization, and Export features	Built interactive dashboard with charts and summary cards; implemented Export to Excel and PDF functionality for reports and transaction logs
Week 7	Future prediction module, system testing, and performance optimization	Developed AI-based future demand prediction module; performed unit and integration testing, resolved functional errors, and optimized database queries
Week 8	Final system integration, documentation, and deployment	Integrated all modules, prepared technical documentation and user guide, conducted final system demo, and completed project submission

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Initial project briefing, requirement analysis, system planning, and technology stack finalization	02-Feb-2026
Database & Backend Setup Completed	Successful design of electronics inventory database schema and implementation of core backend structure with Spring Boot API connectivity	Week-1
Authentication Module Completed	Implementation of secure login system with role-based access control for Admin and Staff users	Week-2
Inventory Module Prototype Ready	Development of core electronics inventory features including item management and stock updates	Week-3
Transaction Tracking & Alert System Completed	Implementation of issue/return logging with automatic stock adjustment, history tracking, and email/SMS notification integration	Week-4
Dashboard, Analytics	Deployment of interactive dashboard with	Week-5

Milestone	Description	Date Achieved
& Export Integration	stock insights, charts, alert indicators, and Export to Excel/PDF feature	
Future Prediction Module Completed	AI-driven future demand forecasting module developed and integrated with historical inventory data	Week-6
System Testing & Optimization Completed	Completion of functional testing, bug fixes, and performance improvements across all modules	Week-7
Final System Deployment & Documentation	Full system integration, preparation of documentation, and successful project demonstration	Week-8

5b. Project Execution Details

The Smart Electronics Inventra Management System was developed using a structured full-stack architecture to ensure scalability, security, and maintainability. The system was divided into key modules including authentication, inventory management, transaction tracking, alert notifications, analytics dashboard, export functionality, and future demand prediction.

The backend was implemented using Java (Spring Boot) to develop RESTful APIs that handle business logic, user requests, and database interactions. A MySQL relational database was used to store and manage system data including user details, electronics inventory records, product categories, transaction logs, and historical usage data. The database schema was designed using normalized tables with proper foreign key relationships to maintain data consistency and integrity.

A secure authentication and authorization system was implemented to protect system access. User credentials were securely stored with password encryption, and role-based access control (RBAC) was enforced to differentiate permissions between Admin and Staff users. Administrators have full control over inventory and user management, while staff members are restricted to operational tasks such as updating stock and recording transactions.

The inventory management module enables users to add new electronics products, edit existing item details, manage stock quantities, assign categories, and perform item searches. Backend validation ensures that stock values remain consistent, and all changes are reflected in real time.

The transaction tracking module was developed to monitor item movement within the system. It records issuance and return operations with detailed information including item ID, quantity, user performing the action, and timestamp. The system automatically updates inventory levels after each transaction, ensuring accurate and up-to-date stock information.

An intelligent alert mechanism was developed to automatically send email and SMS notifications to administrators and managers when stock levels of any electronics item fall below the configured minimum threshold. A scheduled background service in Spring Boot periodically checks stock levels and dispatches these notifications, preventing stockouts and ensuring uninterrupted operational flow.

The system provides comprehensive export functionality allowing users to download inventory lists, transaction histories, and analytics reports in Excel (.xlsx) and PDF formats. This enables offline analysis, sharing with stakeholders, and record-keeping for compliance purposes.

A predictive analytics module was developed using historical transaction data to forecast future stock requirements for electronics items. The module uses trend analysis and moving averages based on historical data stored in MySQL, providing directional forecasts displayed as trend charts on the dashboard to support proactive procurement planning.

The frontend was developed using React.js, providing a responsive and user-friendly interface. The React application communicates with backend APIs to fetch and display inventory data dynamically. An interactive dashboard with charts and summary cards was created to visualize important insights such as total stock, low-stock items, category-wise distribution, and recent transactions.

Testing was conducted at multiple levels including backend API testing, frontend functionality testing, and integration testing between the React frontend and Java backend. Performance was optimized through efficient SQL queries, proper indexing, and API response handling. Finally, all modules were integrated and deployed in a test environment for demonstration. Comprehensive technical documentation and a user guide were prepared to ensure smooth system usage and future scalability.

6. Snapshots / Screenshots

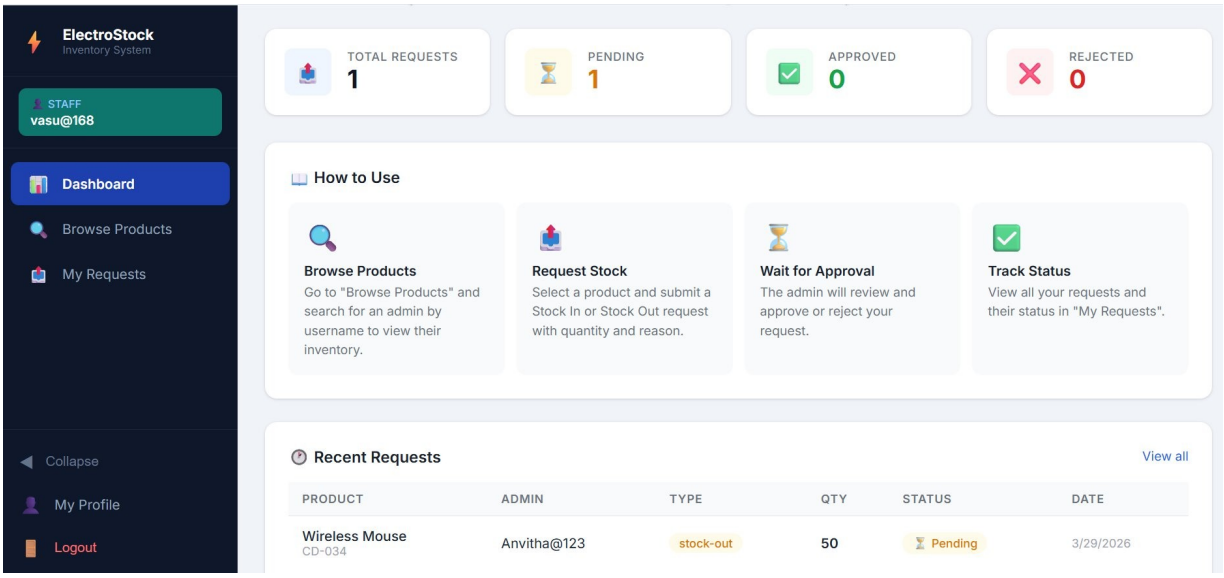

Figure 1: Staff Dashboard – Request Overview and How to Use Guide

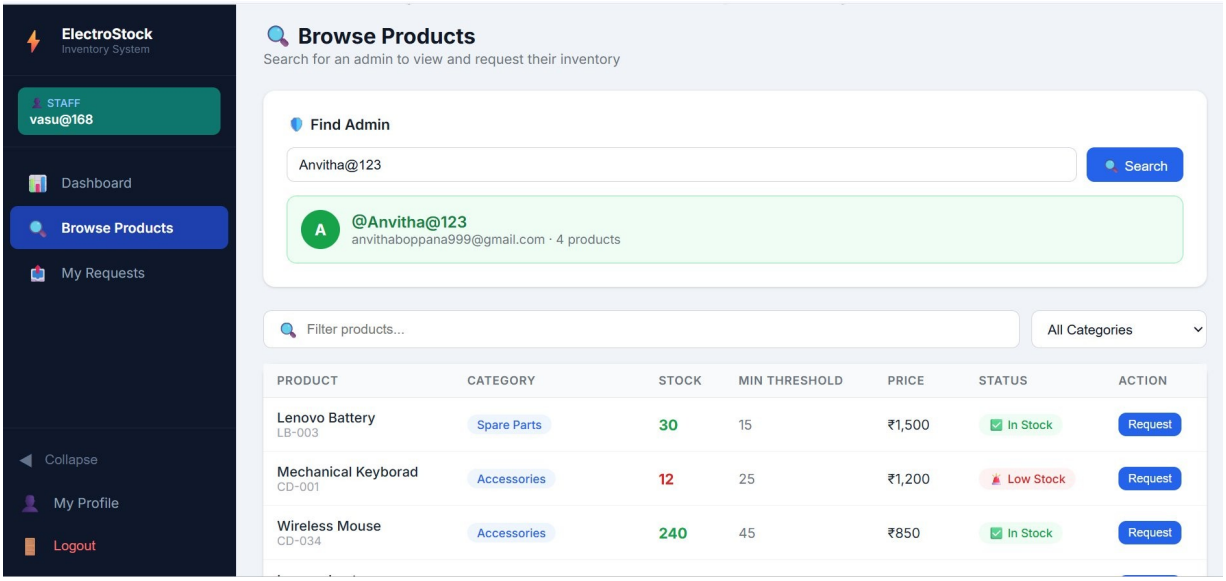

Figure 2: Browse Products – Staff View with Stock Status and Request Option

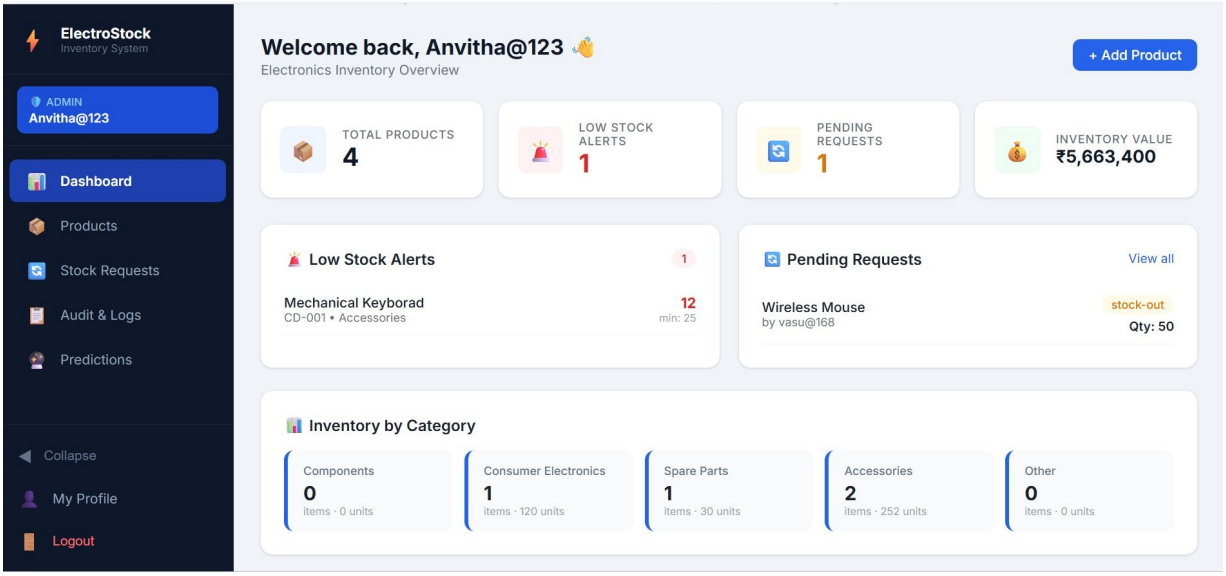

Figure 3: Admin Dashboard – Electronics Inventory Overview with Low Stock Alerts and Category Summary

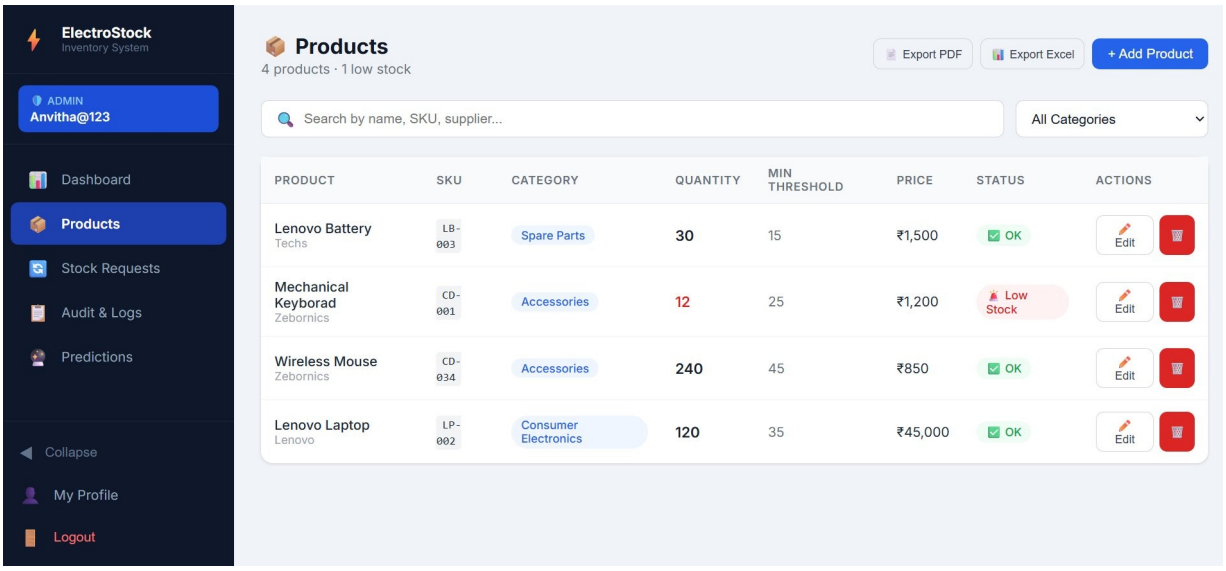

Figure 4: Products Module – Inventory List with Export PDF and Export Excel Options

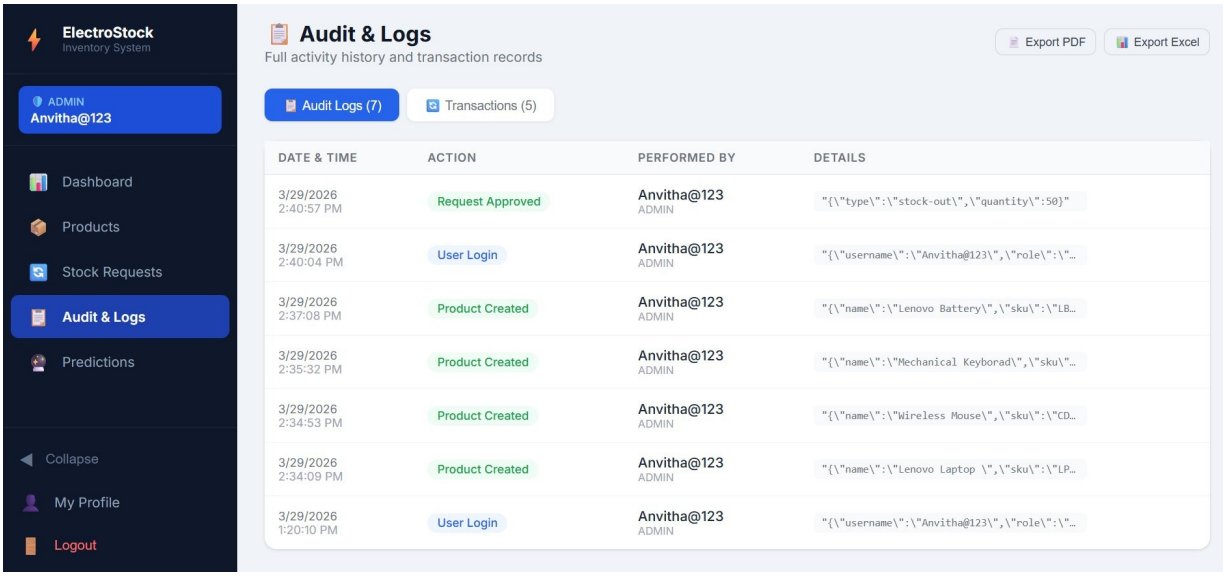

Figure 5: Audit & Logs – Full Activity History and Transaction Records with Export Options

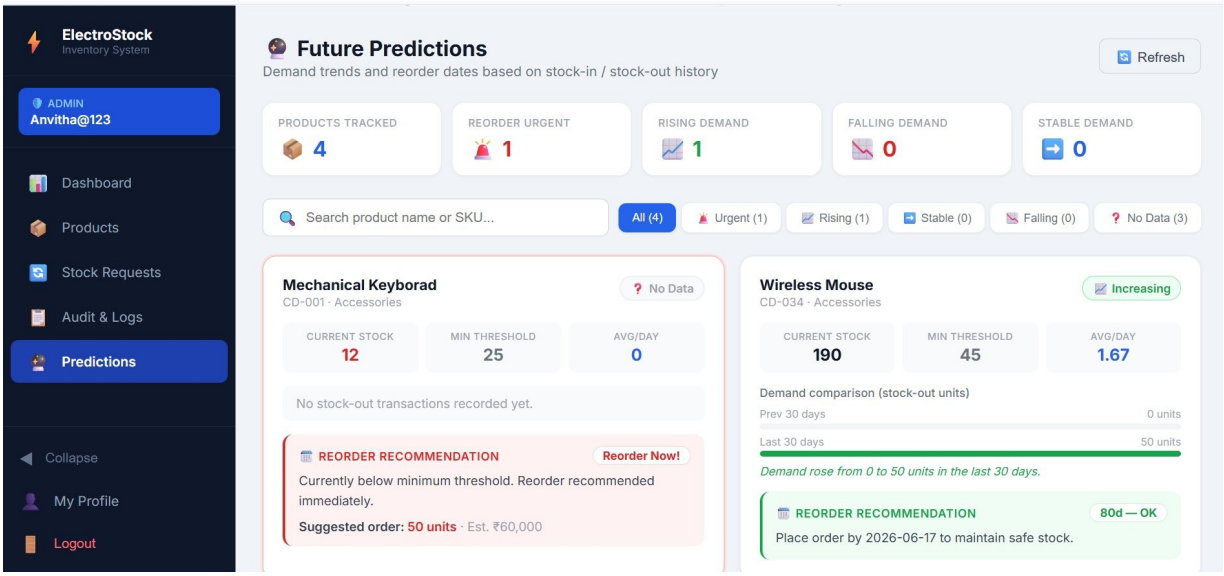

Figure 6: Future Predictions – Demand Trends and Reorder Recommendations based on Stock History

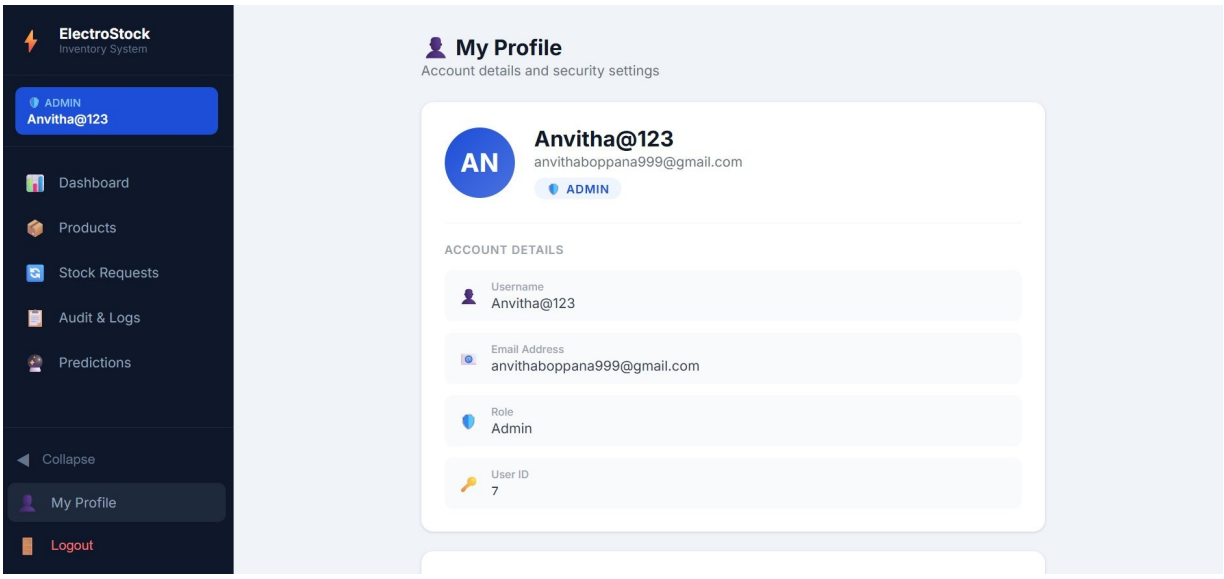

Figure 7: My Profile – Admin Account Details and Security Settings

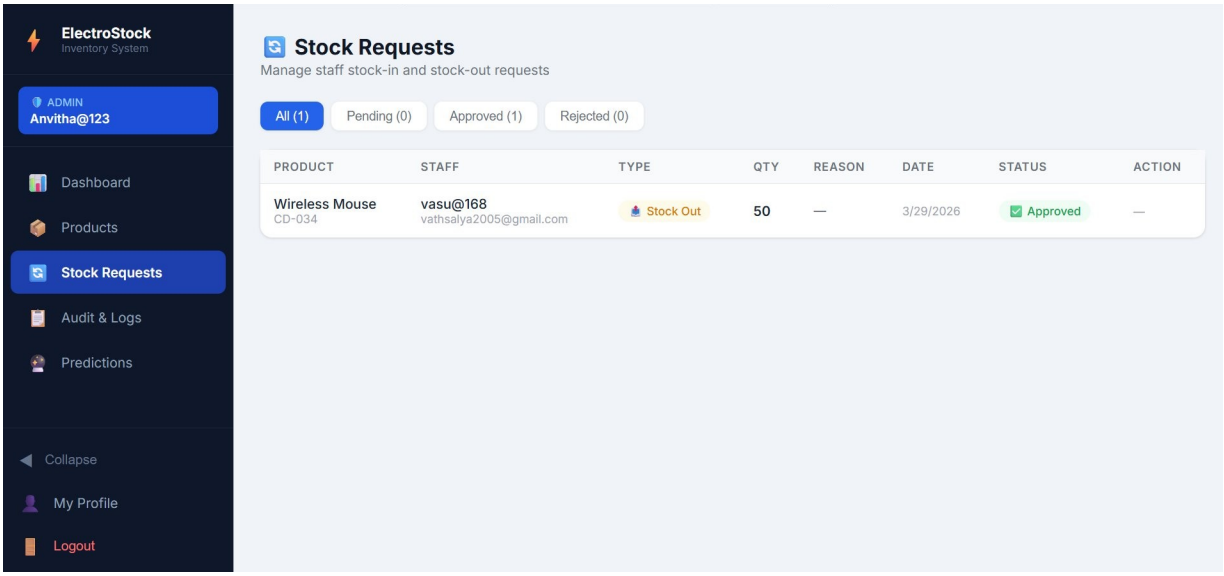

Figure 8: Stock Requests – Admin View of Staff Stock-In and Stock-Out Requests with Approval Status

7. Learnings & Skills Acquired

- Gained practical experience in designing and managing relational databases using MySQL for complex electronics inventory data with proper normalization and indexing
- Developed strong understanding of backend development using Java and Spring Boot for building secure and scalable RESTful APIs
- Learned to implement JWT-based authentication and role-based authorization mechanisms using Spring Security
- Acquired skills in frontend development using React.js for building responsive and dynamic user interfaces
- Improved knowledge of API integration between frontend and backend systems using Axios
- Understood real-world implementation of electronics inventory management logic and stock control systems
- Learned to design and display data visualizations and dashboards for better decision-making
- Gained hands-on experience in implementing email and SMS notification systems for automated alerting
- Developed expertise in implementing export functionality to Excel and PDF formats for business reporting
- Acquired knowledge in predictive analytics and AI-based demand forecasting using historical inventory data
- Enhanced ability in debugging, testing, and optimizing system performance
- Gained experience in full-stack application architecture and module integration
- Improved team collaboration, documentation, and project presentation skills

8. Challenges Face

Challenge 1: Designing an Efficient Database Structure for Electronics

Problem: Managing relationships between electronics product categories, components, suppliers, users, and transactions without data redundancy

Solution: Designed a normalized MySQL schema with proper foreign keys and indexing specifically tailored for electronics inventory to maintain data integrity and improve query performance

Challenge 2: Maintaining Accurate Real-Time Stock Updates

Problem: Ensuring inventory quantities remain correct after every issue and return transaction

Solution: Implemented backend logic in Java Spring Boot to automatically update stock levels and validate quantity constraints

Challenge 3: Implementing the Email and SMS Alert System

Problem: Configuring reliable and timely alert notifications that trigger correctly when stock falls below predefined thresholds without generating excessive false alerts

Solution: Developed a scheduled background service in Spring Boot that periodically checks stock thresholds and dispatches email and SMS notifications to ensure relevant personnel are alerted promptly

Challenge 4: Building the Export to Excel and PDF Feature

Problem: Generating formatted downloadable reports from dynamic inventory data that work consistently across different devices and browsers

Solution: Integrated Apache POI for Excel generation and iText for PDF export in the backend, with the React frontend triggering downloads via API responses with proper content-type headers

Challenge 5: Integrating React Frontend with Java Backend

Problem: Handling smooth data exchange between frontend and backend services

Solution: Built RESTful APIs and implemented proper error handling and response management for seamless integration

Challenge 6: Developing the Future Demand Prediction Module

Problem: Building an accurate prediction model with limited historical transaction data for varied electronics product categories

Solution: Implemented a moving average and trend analysis approach using historical transaction logs stored in MySQL, providing reliable directional forecasts displayed as trend charts on the dashboard

9. Testimonials from Team

Working on the Smart Electronics Inventra Management System was a valuable learning experience for the entire team. Each member contributed to different modules such as backend development, frontend design, database management, alert system integration, and the demand prediction module, which helped us understand how a real-world full-stack system is built and integrated.

Regular discussions, planning sessions, and collaborative debugging helped us overcome technical challenges efficiently. The project improved our practical knowledge in Java, MySQL, and React, while also strengthening our problem-solving abilities. Building features like the email/SMS alert system, export functionality, and the AI-driven demand prediction module gave us exposure to industry-relevant technologies beyond standard development.

This experience enhanced our teamwork, communication, and project documentation skills, making us more confident in handling industry-level software development projects.

10. Conclusion

The Smart Electronics Inventra Management System successfully demonstrates how digital technology can transform traditional electronics inventory handling into a smart, efficient, and data-driven process. By integrating a Java-based backend using Spring Boot, a MySQL relational database, and a React frontend, the system delivers a complete full-stack solution for real-time electronics inventory monitoring and control.

The project effectively addressed common challenges in manual inventory systems such as stock mismatches, lack of transparency, and delayed updates. Through features like role-based authentication, transaction tracking, automated stock updates, email and SMS alerts, Export to Excel and PDF functionality, visual analytics dashboards, and AI-based future demand prediction, the system ensures accuracy, accountability, and ease of management.

Developing this system provided hands-on experience in backend API development, database design, frontend integration, alert system implementation, report generation, and predictive analytics. It also strengthened our understanding of how different components of a software system interact to create a reliable and scalable application.

Overall, the Smart Electronics Inventra Management System stands as a practical and scalable solution that can be adapted for use in electronics retail stores, component warehouses, educational institutions, and corporate IT departments requiring efficient inventory control. The project not only met its technical objectives but also enhanced our readiness to work on real-world software development challenges.

11. Acknowledgements

We would like to express our sincere gratitude to our project mentor and faculty members for their continuous guidance, encouragement, and valuable feedback throughout the development of the Smart Electronics Inventra Management System. Their technical insights and support played a crucial role in helping us overcome challenges and improve the quality of our work.

We also extend our gratitude to Infosys Springboard for providing this Virtual Internship 6.0 platform, which gave us the opportunity to work on a real-world, industry-relevant project. The structured learning environment and project framework greatly facilitated our technical growth.

We thank our institution for providing the necessary resources and learning environment that enabled us to successfully complete this project. The opportunity to work on a real-time, practical system has significantly enhanced our technical knowledge and confidence.

Finally, we appreciate the teamwork, dedication, and collaborative efforts of all team members — Boppana Anvitha, Sathyadev Selvam, Gautam Shivam Mishra, Shaik Firoz, and Aayushi Maurya — whose consistent contributions and commitment made the successful completion of this project possible.